Improving PICOX with Multi-Channel Gated Feature Fusion for Overlapping PICO Entity Extraction

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Abstract. Automatic extraction of Participants, Interventions, Comparisons, and Outcomes (PICO) supports evidence retrieval but remains difficult when entities overlap or have ambiguous boundaries. PICOX handles overlap through type-independent span localization followed by multi-label span classification, yet first-stage boundary errors cannot be recovered downstream. We improve its localization module with multi-channel gated feature fusion. A point channel preserves token semantics, a convolutional window channel captures immediate boundary patterns, and a sentence channel supplies global context. A token-wise gate combines these views while preserving the encoder dimension. Focal loss emphasizes rare boundary labels, and validation-selected start/end thresholds improve recall. On EBM-NLP, the method achieves 58.56% micro F1 and 60.38% macro F1, improving PICOX by 7.69 and 7.61 points. We also construct a Chinese clinical guideline corpus from 721 structured recommendations. With ERNIE-Health, our model obtains 62.37% micro F1, 6.53 points above the Chinese PICOX baseline and 3.19 points above Qwen2.5-72B-Instruct (3-shot). The consistent gains suggest that multi-granularity boundary modeling transfers across biomedical languages and document genres.

Keywords: PICO extraction · Overlapping entities · PubMedBERT · Multi-channel fusion · Focal loss · Chinese clinical guidelines.

1 Introduction

Evidence-based medicine relies on the efficient identification and synthesis of clinical evidence from randomized controlled trial (RCT) publications. Participants, Interventions, Comparisons, and Outcomes (PICO) provide a structured representation of the central clinical questions addressed by an RCT. Automatic PICO extraction can support systematic review screening, evidence retrieval, clinical guideline construction, and downstream evidence synthesis.

Early PICO systems commonly use token-level sequence labeling with conditional random fields (CRFs), recurrent networks, or pretrained language models.

Although effective for flat entities, BIO assigns only one label per token and cannot naturally represent overlap. For example, an outcome measurement may be embedded in a longer participant span.

PICOX [8] addresses this limitation by decomposing PICO extraction into boundary detection and span classification. The boundary detector first predicts whether each token is outside a span, at the start, at the end, simultaneously at both boundaries, or inside a span. Candidate spans are then generated from predicted start–end pairs and classified into PICO categories. This decomposition supports overlapping entities because boundary identification is separated from entity type assignment. However, it also introduces a pipeline bottleneck: if a gold boundary is missed in the first stage, no correct candidate span can be recovered by the second-stage classifier.

The original study reports missed starts, shifted long-span boundaries, and incorrectly merged adjacent entities. Our reproduced error audit further shows that short interventions and outcomes constitute most zero-overlap misses, motivating a boundary model that explicitly combines token, local, and global evidence. Detailed statistics are presented in Sect. 4.

Figure 1 illustrates why localization is the critical failure point through a complete P/I/O extraction example. Stage 1 correctly localizes the participant, misses the intervention, and shifts the outcome start. Candidate generation therefore omits the intervention and contains only an inexact outcome span; Stage 2 can label these candidates but cannot reconstruct the gold boundaries. Under strict evaluation, only the participant is correct. Consequently, first-stage boundary recall imposes an upper bound on end-to-end recall.

To address these problems, we retain the PICOX pipeline, including its multi-label span classifier and composite-span augmentation, but redesign its first-stage boundary detector. We propose a multi-channel gated boundary detector that explicitly represents token, local-window, and sentence-level information. The point channel preserves contextual token features. The window channel applies a one-dimensional convolution to each token and its immediate neighbors to expose short-range boundary patterns. The sentence channel broadcasts `[CLS]` to every token as global PICO context. A learned gate selects the channel mixture for each token. We further introduce focal loss and validation-based adaptive boundary thresholds to improve rare-boundary learning and inference-time recall.

Existing PICOX evaluations focus on English RCT publications. Clinical guidelines provide another important evidence source, but Chinese guideline recommendations differ from English RCT abstracts in language, terminology, sentence structure, and communicative purpose. To test whether the proposed boundary improvements capture general PICO structure rather than English-specific patterns, we additionally construct a Chinese guideline PICO corpus from structured recommendation records and replace PubMedBERT with the Chinese medical encoder ERNIE-Health. This creates two complementary evaluation settings: EBM-NLP measures improvements on the original overlapping

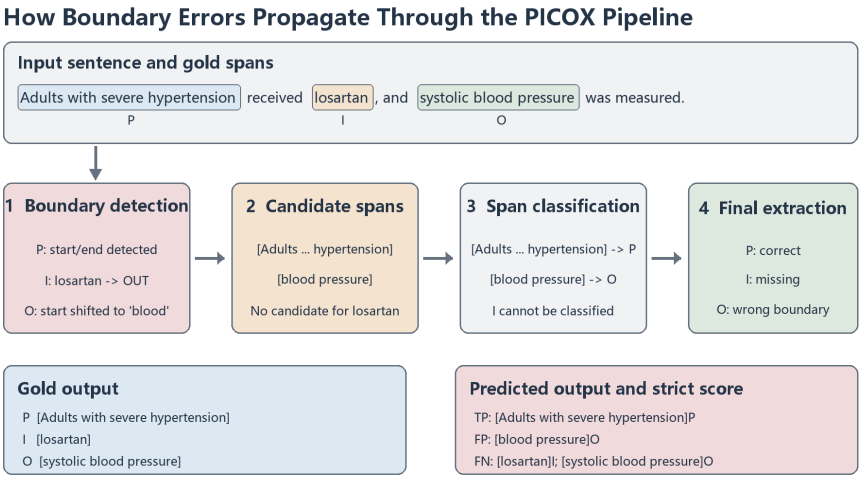


Fig. 1. Complete P/I/O extraction example in the two-stage PICOX pipeline. A missed intervention and shifted outcome boundary propagate through candidate generation and cannot be repaired by span classification.

PICO benchmark, while the Chinese corpus measures transfer of the architecture across both language and document genre.

Relative to the original PICOX framework, the main contributions of this work are as follows:

1. We improve PICOX's boundary detector with token-wise gated point, window, and sentence features, focal loss, and separately calibrated start/end thresholds.
2. We validate the method on English RCT abstracts and a new Chinese guideline corpus, improving micro F1 over PICOX by 7.69 and 6.53 points.

Code and data availability. The implementation, documentation, and dataset access instructions are available at `https://github.com/gongrui999/mc-picox`.

2 Related Work

2.1 PICO Entity Extraction

The EBM-NLP corpus introduced large-scale token-level PICO annotations for RCT abstracts and has become a standard benchmark for evidence extraction [1]. Earlier approaches employed CRFs and recurrent neural networks, while later studies adopted contextual representations from BioBERT [2], SciBERT [3], PubMedBERT [4], and BioELECTRA [5]. Domain-specific pretraining improves

biomedical terminology modeling, but a conventional sequence-labeling output layer still cannot represent multiple overlapping spans at one token position.

This limitation is structural rather than merely a consequence of encoder quality. BIO-style models jointly encode entity type and boundary position in one tag, so a token shared by two spans would require two incompatible tags. Layered or multi-head tagging can partially address selected overlap patterns, but it requires additional label structures and becomes difficult when the number or configuration of overlaps is not fixed. General span-enumeration models avoid this conflict by classifying substrings directly; however, they usually impose a maximum span width and increase the number of candidates as the sequence grows. These trade-offs motivate a boundary-driven span formulation for variable-length overlapping PICO entities.

PICOX reformulates PICO extraction as a two-stage span-based task [8]. It first detects type-independent boundaries and subsequently assigns one or more PICO types to each candidate span. This framework achieves stronger performance than token-level baselines on EBM-NLP and supports overlapping entities without imposing a maximum entity length. It additionally constructs composite spans by crossing boundaries of different gold entities and uses them as hard negative examples for the span classifier. Its ablation study shows that this augmentation primarily reduces false positives.

PICOX therefore solves overlap representation and candidate discrimination, but it does not remove dependence on first-stage recall. Composite-span augmentation teaches the second stage to reject implausible candidates; it cannot create a candidate whose start or end was not proposed. Moreover, the localizer uses one contextual token vector, an unweighted objective, and a shared boundary threshold. The literature consequently leaves a specific gap: improving type-independent boundary recall without discarding PICOX's variable-length span construction and multi-label classifier. Our work targets this gap; the component-level differences are detailed in Sect. 3.2.

Recent generative approaches provide a complementary direction. AlpaPICO uses in-context learning and parameter-efficient instruction tuning for PICO frame extraction [9], while large-scale prompting has been used to extract PICOs from hundreds of thousands of trial abstracts [10]. LLMs have also been evaluated for numerical trial-result extraction [13], end-to-end evidence synthesis [14], and full-text RCT data extraction [16]. Open-weight instruction models such as Qwen2.5 [15] make few-shot generative extraction reproducible without a proprietary API. These studies broaden evidence automation, but they also report sensitivity to output format, complex outcomes, and verification requirements. Exact, reproducible span boundaries therefore remain important for corpus-based PICO extraction.

2.2 Multi-Granularity Feature Fusion

Transformer encoders contextualize each token across the sequence, but a shallow classification head may not sufficiently emphasize explicit local patterns. Recent nested NER research likewise combines span reasoning with multi-granularity or

task-adaptive representations in medical text [11], while LLM-based nested NER studies show that explicit structural instructions remain necessary for reliable overlap extraction [12]. Convolutional local features and a sentence representation can therefore complement the implicit contextualization of a Transformer.

Simple concatenation treats all feature sources equally and increases the input dimension of the classifier. In contrast, gated fusion learns input-dependent weights. This is important for PICO boundary detection because different tokens require different evidence: an unambiguous biomedical term may rely mainly on the point channel, a short ambiguous entity may require local trigger words, and a section-dependent expression may benefit from sentence-level semantics.

Existing contextual encoders and explicit local modules should not be viewed as substitutes. Self-attention supplies broad semantic interaction, whereas a small convolution imposes a useful locality bias for patterns such as "primary endpoint was" or coordinated entity boundaries. Likewise, a global sentence vector can indicate whether an otherwise generic term occurs in a population, treatment, or outcome context. The unresolved design question is how to combine these signals: fixed averaging cannot adapt across tokens, and concatenation enlarges the classifier while leaving it to infer channel importance implicitly. We therefore use a normalized token-wise gate, which retains the encoder dimension and exposes the selected evidence at every position.

2.3 Learning from Imbalanced Boundary Labels

In boundary detection, the OUT class is much more frequent than START, END, and BOTH. Standard cross-entropy can be dominated by easy outside tokens, encouraging conservative predictions. Focal loss down-weights easy examples and concentrates learning on difficult instances [6]. This property is especially suitable for the rare BOTH class, which represents one-token entities and simultaneously provides both start and end evidence.

Boundary imbalance affects both optimization and decoding. During training, the model can obtain low average loss by correctly classifying abundant, easy OUT tokens while under-learning rare starts, ends, and one-token entities. During inference, a single threshold assumes that start and end scores have the same calibration even though their frequencies and contextual cues differ. Class weighting addresses frequency but treats easy and difficult examples within a class similarly; threshold tuning changes decisions but cannot repair a poorly trained representation. Focal loss and separate validation-selected start/end thresholds are therefore complementary: the former changes where gradients are concentrated, while the latter calibrates the resulting evidence without using test labels.

2.4 Cross-Language and Guideline PICO Extraction

Most benchmarked PICO extraction systems are developed for English RCT abstracts. Chinese guideline recommendations differ not only in language but also

in communicative function: they often compress population restrictions, treatment actions, and numeric targets into one recommendation sentence. Domain-specific Chinese encoders such as ERNIE-Health [7] improve terminology representation, but changing the encoder alone does not test whether the boundary architecture generalizes. There is also a data gap because structured guideline fields are not directly equivalent to textual spans; paraphrases and combined indicator–target expressions require alignment and boundary review. We address this gap by constructing CCG-PICO and evaluating the same localization design with a language-appropriate encoder rather than claiming zero-shot transfer.

The preceding subsections discuss these formulations in detail. Table 1 provides a compact comparison. Our contribution is a recall-oriented redesign of PICOX localization, evaluated across language and document genre.

Table 1. Compact comparison of PICO extraction formulations.

Approach	Extraction unit	Decoding	Key property
BIO-based PLM/CRF	Token tag	Linear or CRF layer	Efficient, but cannot encode arbitrary overlap.
General span enumeration	Candidate span	Enumerate and score spans	Supports overlap; candidate cost grows rapidly.
PICOX	Boundary pair and span	Boundary pairing, then multi-label classification	Supports variable-length overlap; boundary errors are irreversible.
Few-shot LLM	Generated span	Prompted generation	Flexible, but prompt- and alignment-sensitive.
Ours	PICOX boundary pair and span	**Gated multi-channel localization, focal loss, and calibrated thresholds**	**Recall-focused; evaluated across language and genre.**

3 Method

This section describes how the proposed model extends the first stage of PICOX while leaving its multi-label span classifier and composite-span augmentation unchanged. We first formulate type-independent boundary localization and summarize the original PICOX computation. We then introduce the shared biomedical encoder, three parallel feature channels, token-wise gated fusion, focal-loss optimization, and validation-calibrated boundary decoding. The complete workflow is therefore *encoding* $\rightarrow$ *multi-granularity boundary representation* $\rightarrow$ *gated fusion* $\rightarrow$ *boundary decoding* $\rightarrow$ *span classification*. This organization makes clear

which components are inherited from PICOX and which are introduced in this work.

3.1 Task Formulation

Let a tokenized sentence be $X = (x_1, \ldots, x_n)$. The boundary detector assigns one of five labels to each token:

$$\mathcal{Y}_b = \{\textsc{Out}, \textsc{Start}, \textsc{End}, \textsc{Both}, \textsc{In}\}.$$

Both denotes a position that is simultaneously a start and an end, which commonly corresponds to a one-token entity. A start position s and a compatible end position e form a candidate span (s, e). The second-stage classifier assigns zero or more labels from Participants, Interventions, and Outcomes to each candidate. Comparisons are included in the intervention class, following the EBM-NLP annotation scheme.

3.2 Original PICOX Baseline and Our Extension

The original PICOX localization module first obtains a contextual vector h_i for token x_i and directly computes the distribution over five relative positions:

$$p_i^b = \text{softmax}(W_b h_i + b_b). \tag{1}$$

It is trained with standard cross-entropy. At inference, a token is treated as a possible boundary when the probability of Start, End, or Both exceeds a shared empirical threshold t. All legal start–end pairs are enumerated, after which a separate PubMedBERT classifier assigns P/I/O labels using sigmoid outputs. Composite spans generated from mismatched gold boundaries are added as non-entity training examples.

This design successfully resolves two limitations of conventional PICO sequence labeling: overlapping entities can be represented, and no maximum span width needs to be assumed. Nevertheless, the linear localization head, unweighted loss, and shared fixed threshold leave three opportunities for improvement. First, the decision layer does not explicitly separate lexical, local-transition, and sentence-level evidence. Second, the large number of Out tokens can dominate optimization. Third, one threshold cannot independently calibrate start and end evidence. These limitations motivate a targeted redesign of localization while preserving PICOX's span classifier and hard-negative augmentation.

Figure 2 presents the complete architecture and the division between the two stages. In the redesigned first stage, a shared domain-specific encoder supplies point, convolutional-window, and sentence features. Token-wise gated fusion feeds a five-label boundary head trained with focal loss, and separate start/end thresholds produce boundary positions.

The retained second stage enumerates legal start–end pairs and assigns independent P/I/O probabilities with the original sigmoid classifier, allowing overlapping and multi-label outputs. Composite spans remain hard negatives. Thus, all new components act before candidate generation; PICOX's span-level decision mechanism is unchanged.

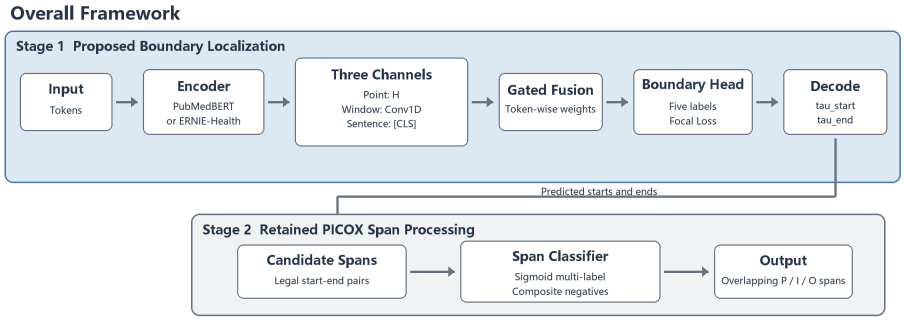


Fig. 2. Overall framework. Stage 1 is the proposed boundary localizer; Stage 2 retains PICOX's candidate generation and multi-label span classification.

3.3 PubMedBERT Encoder

We use PubMedBERT as the base encoder because it is pretrained from scratch on biomedical literature and provides domain-specific lexical representations. For a batch of token sequences, the encoder produces

$$H = \text{PubMedBERT}(X) \in \mathbb{R}^{B \times L \times d}, \tag{2}$$

where B is the batch size, L is the padded sequence length, and d is the backbone hidden dimension. The same tensor H is shared by all three feature channels, so PubMedBERT is executed only once.

3.4 Three Parallel Feature Channels

Point Channel. The point channel directly preserves the contextual representation of every token:

$$V^p = H. \tag{3}$$

Although every vector in H already contains contextual information through self-attention, this channel retains the strongest token-specific biomedical semantics. It is useful for explicit drug names, diseases, and clinical measurements whose lexical identity is highly informative.

Window Channel. Boundary decisions often depend on short-range patterns. For instance, "primary endpoint was" strongly indicates that a following short expression may be an outcome, whereas "patients receiving" often precedes an intervention. We therefore apply a one-dimensional convolution over the token sequence:

$$V^w = \text{Conv1D}(H^\top)^\top, \tag{4}$$

where the convolution has d input channels, d output channels, kernel size 3, and padding 1. Consequently, the representation at position i explicitly combines (h_{i-1}, h_i, h_{i+1}). Kernel size 3 is selected because it captures the immediate

left and right contexts most directly associated with boundary transitions while limiting noise and computational overhead. Padding preserves the original sequence length.

The window channel is not equivalent to averaging adjacent vectors. Its position-specific convolution weights allow the model to learn that a left trigger and a right continuation have different effects on a boundary decision. It is particularly helpful for one- to three-token entities, for which a one-token boundary shift causes an entirely incorrect strict span.

Sentence Channel. Some expressions cannot be disambiguated from their immediate neighbors alone. For example, "risk" can denote an outcome in a result sentence but can also occur as an ordinary modifier in a participant description. We use the `[CLS]` output h_{cls} as a global sentence representation and broadcast it to every position:

$$V_i^s = h_{\text{cls}}, \qquad i = 1, \ldots, L. \tag{5}$$

Thus, $V^s \in \mathbb{R}^{B \times L \times d}$. The channel informs each token whether the complete sentence mainly describes enrollment criteria, interventions, or clinical outcomes. Broadcasting does not allocate additional memory for independent copies when implemented with tensor expansion.

3.5 Token-Wise Gated Fusion

The relative usefulness of the three channels varies across positions. We first concatenate the channel vectors:

$$Z_i = [V_i^p; V_i^w; V_i^s] \in \mathbb{R}^{3d}. \tag{6}$$

A linear layer followed by softmax produces three normalized weights:

$$G_i = \text{softmax}(W_g Z_i + b_g) = [g_i^p, g_i^w, g_i^s], \tag{7}$$

where $W_g \in \mathbb{R}^{3 \times 3d}$ and $g_i^p + g_i^w + g_i^s = 1$. The fused representation is

$$H_i^f = g_i^p V_i^p + g_i^w V_i^w + g_i^s V_i^s. \tag{8}$$

Equation 8 keeps the output dimension equal to d, unlike direct concatenation, which would increase the classification input to $3d$. The gate is learned jointly with the complete network. For a short ambiguous expression, the model can assign a larger weight to V^w; for a distinctive biomedical term, it can emphasize V^p; and for a section-dependent expression, it can increase V^s. The returned gate weights also provide an interpretable view of the model's decision process.

3.6 Boundary Classification and Focal Loss

The fused representation is passed through dropout and a linear classifier:

$$z_i = W_c \,\text{Dropout}(H_i^f) + b_c, \tag{9}$$

where $z_i \in \mathbb{R}^5$. To reduce the dominance of easy outside tokens, we use class-weighted focal loss:

$$\mathcal{L}_{\text{focal}} = -\frac{1}{N}\sum_{i=1}^{N} \alpha_{y_i}(1 - p_{i,y_i})^{\gamma} \log p_{i,y_i}. \tag{10}$$

Here p_{i,y_i} is the predicted probability of the gold label, $\gamma = 2$, and the label weights for (OUT, START, END, BOTH, IN) are initialized as $(0.15, 2.0, 2.0, 6.0, 1.0)$. The high weight of BOTH specifically strengthens supervision for one-token spans. These weights are selected using the validation split and remain fixed for test evaluation.

3.7 Adaptive Boundary Decoding

At inference time, START and BOTH both provide start evidence, whereas END and BOTH provide end evidence. We define

$$q_i^s = p_i(\text{START}) + p_i(\text{BOTH}), \tag{11}$$
$$q_i^e = p_i(\text{END}) + p_i(\text{BOTH}). \tag{12}$$

A token is selected as a start if $q_i^s > \tau_s$ and as an end if $q_i^e > \tau_e$. Rather than adopting one fixed empirical threshold for both decisions, (τ_s, τ_e) is selected by grid search on the validation set. The test set is used only once after threshold selection. This strategy recovers low-confidence but informative boundary signals while avoiding direct optimization on the test data.

3.8 Span Classification and Post-processing

All valid start–end pairs with $s \leq e$ are enumerated and passed to the second-stage PubMedBERT span classifier. Independent sigmoid outputs support multi-label classification, allowing one span to receive more than one PICO type. During training, incorrectly crossed boundaries from different gold entities are included as hard negative spans. Finally, non-maximum suppression removes redundant same-type predictions according to textual overlap and confidence.

4 Experiments

The experiment code, configurations, and dataset instructions are publicly available online.[3]

[3] `https://github.com/gongrui999/mc-picox`

4.1 Dataset

EBM-NLP. We evaluate the proposed model on EBM-NLP [1]. It contains RCT abstracts with token-level annotations for Participants (P), Interventions and Comparisons (I), and Outcomes (O). Table 2 reports the number of entity annotations. The test set contains 643 participant, 1,726 intervention, and 1,833 outcome spans. Approximately 8.2% of the entities overlap, making the corpus suitable for evaluating span-based extraction.

Table 2. Number of entity annotations in EBM-NLP.

Split	Participants	Interventions	Outcomes	Total
Training	17,952	32,859	33,554	84,365
Test	643	1,726	1,833	4,202
All	18,595	34,585	35,387	88,567

Chinese Clinical Guideline PICO Corpus. To examine whether the proposed localization architecture transfers beyond English RCT abstracts, we construct a Chinese Clinical Guideline PICO corpus, referred to as CCG-PICO, from 721 guideline recommendations. Each source record contains a normalized recommendation sentence and structured fields describing disease, subtype, age, gender, other demographic constraints, comorbidities, diagnostic criteria, intervention category, intervention name, intervention details, outcome indicator, and outcome target.

The structured fields are aligned with their textual realizations in the final recommendation. Disease, age, gender, demographic constraints, and comorbidities form participant candidates; intervention names and details form intervention candidates; and outcome indicators and targets form outcome candidates. Exact string matching is first used to obtain high-confidence spans. For structured values that are paraphrased in the recommendation, such as "sodium and potassium management" versus "reduce sodium intake and increase potassium intake", candidate spans are aligned at the token level and manually corrected. All aligned boundaries are subsequently reviewed in the complete sentence context.

The corpus contains participant information in all 721 records, intervention information in 678 records (94.0%), and outcome information in 547 records (75.9%). The average normalized recommendation length is 86 Chinese characters, with a maximum of 186. The control field is empty in the current collection; therefore, as in EBM-NLP, the reported task uses three output categories, P, I, and O. We divide the corpus into 541 training, 90 validation, and 90 test records using a fixed stratified split. Recommendations originating from the same normalized template are assigned to only one split to reduce template leakage.

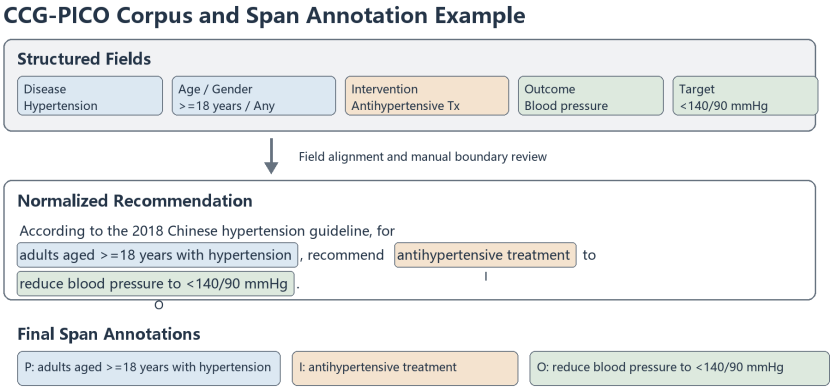


Fig. 3. Example of structured-field alignment and P/I/O span annotation in CCG-PICO. Blue, orange, and green denote participant, intervention, and outcome spans, respectively.

Figure 3 illustrates the construction process with one hypertension recommendation. Structured population, intervention, outcome, and target fields are first associated with their textual realizations. The aligned phrases are then reviewed in sentence context and converted into strict P/I/O spans. The example also shows why the structured fields cannot be used as labels without alignment: the final outcome span combines the indicator "blood pressure", the action "reduce to", and the target value.

4.2 Baselines

We compare with four biomedical pretrained encoders used as token-level sequence-labeling baselines: BioELECTRA, SciBERT, BioBERT, and PubMedBERT. We also compare with PICOX, the span-based baseline upon which our method is built. The proposed method, denoted MC-Gate+FL+AT, adds multi-channel gated fusion (MC-Gate), focal loss (FL), and adaptive thresholding (AT) to the PICOX boundary detector while retaining the same second-stage span classifier.

To include a high-capacity generative baseline, we additionally evaluate the official Qwen2.5-72B-Instruct model [15] on both corpora using three-shot in-context learning. For each corpus, the model receives three fixed examples from its training split followed by one test sentence and returns the exact text and P/I/O type of every span in JSON format. No model parameters are updated. We denote this baseline Qwen2.5-72B-Instruct (3-shot).

This Qwen experiment is a strong high-capacity generative reference and does not imply task-specific optimality. It also represents a different supervision regime: Qwen receives only three demonstrations at inference, whereas the sequence-labeling and PICOX models are optimized on the complete training

set. The comparison therefore evaluates two practical extraction paradigms, not data-matched training efficiency.

For CCG-PICO, we compare two families of Chinese baselines. BERT-Chinese with a sequence-labeling head represents a general-domain Chinese encoder. ERNIE-Health [7] with the same head evaluates the benefit of Chinese medical-domain pretraining. We then replace PubMedBERT in PICOX with ERNIE-Health to form a Chinese PICOX baseline. Qwen2.5-72B-Instruct provides a three-shot generative comparison. Finally, ERNIE-Health+Ours adds the same multi-channel localization, focal loss, and adaptive threshold components used in the English experiment. This design separates gains from the Chinese medical encoder, the extraction framework, and the proposed boundary architecture.

4.3 Evaluation Metrics

We adopt strict span-level evaluation. A predicted entity is counted as a true positive only when its category, start position, and end position exactly match a gold span. Precision, recall, and F1 are defined as

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F1 = \frac{2PR}{P + R}. \tag{13}$$

Micro F1 aggregates true positives, false positives, and false negatives over all entity types. Macro F1 averages the F1 values of P, I, and O.

4.4 Implementation Details

Main settings.

- **Backbone:** PubMedBERT-large, hidden size $d = 1024$.
- **Window channel:** Conv1d, kernel size 3, padding 1, d input/output channels; dropout 0.1.
- **Optimization:** AdamW, learning rate 5×10^{-5}, weight decay 0.01, batch size 8, and 8 epochs.
- **Boundary learning:** focal-loss $\gamma = 2$; separate start/end thresholds searched from 0.05 to 0.30 in steps of 0.01.
- **Model selection:** highest validation boundary recall.

All PICOX variants use identical data splits, tokenization, span classification, and post-processing.

Qwen baseline. Qwen2.5-72B-Instruct is accessed through the Alibaba Cloud ModelScope API. The three demonstrations for each corpus are manually selected from its training split to cover P, I, O, short, and overlapping spans. Every test instance uses the corresponding fixed demonstrations and deterministic greedy decoding. The model returns `text` and `type` in JSON; exact matching maps strings to token offsets, discards unmatched outputs, and merges duplicates. No validation or test examples are used as demonstrations. On EBM-NLP,

strict evaluation yields 2,000 TP, 1,185 FP, and 2,202 FN (62.79% precision, 47.60% recall, and 54.15% micro F1).

Chinese setting. For Chinese experiments, PubMedBERT and its tokenizer are replaced by ERNIE-Health, while the three-channel structure and all output definitions remain unchanged. Character-level boundaries are projected onto subwords. Thresholds are tuned only on the validation set, and all baselines use the same 541/90/90 split.

4.5 Results Across English and Chinese Corpora

Table 3 presents category-level and aggregate results for both evaluation settings in one comparison. This unified view separates three sources of improvement: domain-specific pretraining, the span-based PICOX formulation, and the proposed localization enhancements.

Table 3. Performance across EBM-NLP and CCG-PICO (%). SL denotes sequence labeling.

Dataset	Method	P-F1	I-F1	O-F1	Micro F1	Macro F1
EBM-NLP	BioELECTRA-SL	57.46	46.70	38.47	45.25	47.54
	SciBERT-SL	58.79	46.03	38.21	45.07	47.68
	BioBERT-SL	57.99	47.57	38.01	45.45	47.85
	PubMedBERT-SL	59.91	45.92	37.81	45.05	47.88
	PICOX	60.85	54.68	42.77	50.87	52.77
	Qwen2.5-72B (3-shot)	63.12	57.31	47.96	54.15	56.13
	Ours	**66.74**	**61.03**	**53.36**	**58.56**	**60.38**
CCG-PICO	BERT-Chinese-SL	56.84	47.21	43.56	48.12	49.20
	ERNIE-Health-SL	61.37	51.85	47.92	52.63	53.71
	ERNIE-Health+PICOX	63.25	56.48	51.16	55.84	56.96
	Qwen2.5-72B (3-shot)	66.85	60.92	53.74	59.18	60.50
	ERNIE-Health+Ours	**69.15**	**63.27**	**58.43**	**62.37**	**63.62**

EBM-NLP. PICOX obtains 50.87% micro F1, 5.42 points above the strongest sequence-labeling micro F1. Qwen2.5-72B-Instruct with three-shot prompting obtains 54.15% micro F1, 9.10 and 3.28 points above PubMedBERT-SL and PICOX, respectively. Its 62.79% precision and 47.60% recall (2,000 TP, 1,185 FP, and 2,202 FN) show that the three-shot model is comparatively conservative: the main remaining limitation is missed gold spans rather than excessive generation. Our model further reaches 58.56% micro F1 and 60.38% macro F1, outperforming the 72B generative baseline by 4.41 micro-F1 points. The proposed model's overall precision and recall are 56.13% and 61.24%, compared with 53.49% and 48.50% for PICOX. These differences should be interpreted with the supervision

distinction in mind: Qwen measures three-shot in-context extraction, whereas the other models use full supervised training.

The gains are 5.89, 6.35, and 10.59 points for participants, interventions, and outcomes. Outcomes benefit most because they contain many short, abstract, and context-dependent expressions. Local convolution strengthens short-range triggers, while focal loss and calibrated thresholds reduce the tendency to assign OUT to weak boundaries.

CCG-PICO. BERT-Chinese-SL obtains 48.12% micro F1. Medical-domain pretraining with ERNIE-Health raises it to 52.63%, and the original PICOX formulation further raises it to 55.84%. Qwen2.5-72B-Instruct (3-shot) reaches 59.18% micro F1 and 60.50% macro F1, exceeding ERNIE-Health+PICOX by 3.34 micro-F1 points. The complete method achieves 62.37% micro F1 and 63.62% macro F1, gains of 6.53 and 6.66 points over ERNIE-Health+PICOX and 3.19 micro-F1 points over Qwen.

Participant, intervention, and outcome F1 increase by 5.90, 6.79, and 7.27 points over ERNIE-Health+PICOX, respectively. Relative to Qwen, the gains are 2.30, 2.35, and 4.69 points. Participant descriptions are comparatively regular because age, disease, and comorbidity are often explicit. Intervention and outcome expressions contain more paraphrases, coordinated drugs, target values, and compact clinical phrases, making local-window and sentence-level features particularly useful.

4.6 Joint Ablation Study

Table 4 places the two ablation series in one table. Multi-channel fusion provides consistent initial gains on both corpora. Focal loss yields a larger second gain, confirming that rare boundary labels remain important after changing language and encoder. Adaptive thresholds produce the strongest recall-oriented final configuration.

Table 4. Cumulative ablation results on both corpora (%). MC: multi-channel gated fusion; FL: focal loss; AT: adaptive thresholds.

Dataset	MC	FL	AT	Micro F1	ΔF1
EBM-NLP	–	–	–	50.87	–
	✓	–	–	52.57	+1.70
	✓	✓	–	55.63	+4.76
	✓	✓	✓	58.56	+7.69
CCG-PICO	–	–	–	55.84	–
	✓	–	–	57.23	+1.39
	✓	✓	–	59.86	+4.02
	✓	✓	✓	62.37	+6.53

On EBM-NLP, adding MC, FL, and AT contributes cumulative gains of 1.70, 4.76, and 7.69 points. The corresponding gains on CCG-PICO are 1.39, 4.02, and 6.53 points. The slightly smaller Chinese gain is consistent with the smaller corpus and the more regular structure of normalized recommendation sentences. The matching trend across both corpora supports the complementarity of the three components.

4.7 Cross-Language and Cross-Genre Discussion

The Chinese experiment is not zero-shot transfer from PubMedBERT; it tests the same localization architecture after replacing the language-specific encoder and changing the genre from RCT abstracts to guideline recommendations. Consistent improvements over both PICOX baselines suggest that the model captures general boundary evidence rather than English-specific surface patterns. In Chinese text, the point channel represents diseases, drugs, and measurements; the convolutional channel captures recommendation patterns such as "recommended use" and "should be controlled below"; and the sentence channel distinguishes diagnostic criteria, treatment actions, and outcome targets.

4.8 Discussion on the Complementary Properties of Three Channels

To explain the model behavior, we consider three representative boundary conditions. First, distinctive terms such as drug names can often be identified from their contextual token vectors, so the gate can emphasize the point channel. Second, a short ambiguous expression such as "efficacy" becomes more informative when its immediate context contains "primary endpoint". The window channel explicitly models this pattern and helps distinguish BOTH from OUT. Third, expressions such as "risk" or "response" may have different roles in methods and results sentences. The sentence channel supplies global semantics that help resolve this ambiguity.

The softmax gate is preferable to a fixed average because the useful context varies by token. It is also preferable to pure concatenation because it keeps the representation at d dimensions and limits the number of additional parameters. For hidden size d, the convolution introduces approximately $3d^2$ weights and the gate introduces $9d$ weights. Relative to the PubMedBERT-large encoder, the additional computational cost remains modest.

4.9 Error Analysis

We separate errors inherited from PICOX from those that remain after our modifications. We first audit the 4,202 EBM-NLP gold spans predicted by the reproduced PICOX baseline, and then compare its failures with the outputs and aggregate behavior of the proposed model. As shown in Table 5, 2,152 spans (51.2%) are strictly correct. Another 347 (8.3%) and 341 (8.1%) have their

nearest candidate boundary displaced by one and two tokens, respectively. The remaining 1,362 spans (32.4%) have no exact recovery within this two-token boundary neighborhood. The system additionally produces 809 false-positive spans. Thus, 16.4% of gold entities are close to correct but fail strict matching, whereas almost one third require more than a small boundary correction.

Table 5. Boundary-distance audit of 4,202 EBM-NLP gold spans under the reproduced PICOX baseline.

Outcome for a gold span	Count	Percentage
Strictly correct	2,152	51.2
Nearest boundary offset by 1 token	347	8.3
Nearest boundary offset by 2 tokens	341	8.1
No exact recovery within 2 tokens	1,362	32.4

We then apply a stricter zero-overlap criterion to the missed predictions. This identifies 834 gold spans with no overlapping predicted entity. Table 6 shows that 785 of these cases (94.1%) are interventions or outcomes, and 658 (78.9%) contain only one to three tokens. Among the 658 short misses, 403 (61.2%) never occur as an exact gold-span string in the training set, indicating substantial lexical variation. Nevertheless, 475 (72.2%) have at least one predicted boundary within three tokens, suggesting that the encoder often detects a relevant region but fails to place or pair the exact boundaries.

Table 6. Composition of the 834 zero-overlap missed spans.

Error subset	Count	Percentage
Participants	49	5.9
Interventions	371	44.5
Outcomes	414	49.6
One-token spans	307	36.8
Two-token spans	228	27.3
Three-token spans	123	14.7
Four or more tokens	176	21.1

Compared with PICOX, the proposed model raises recall from 48.50% to 61.24% and precision from 53.49% to 56.13%, yielding a 7.69-point micro-F1 gain. The 12.74-point recall increase shows that the main benefit is recovery of missed candidates, while the P/I/O F1 gains of 5.89/6.35/10.59 points indicate the largest improvement for outcomes. Focal loss and calibrated thresholds recover weak boundaries, the window channel sharpens short local patterns, and the sentence channel disambiguates generic clinical terms.

Figure 4 relates three PICOX failures to their status under our model. The missed one-token outcome "efficacy" is directly targeted by local features, focal loss, and threshold calibration. The merged outcomes "epistaxis" and "headache" are reduced by local transition evidence and hard negatives, but independent start–end pairing can still produce this error. The partially localized 33-token outcome remains unresolved because its boundaries exceed the kernel-size-three window. More generally, the model still cannot recover a boundary absent from its candidate set; long-span consistency, adjacent cross-pairing, and occasional short false positives from lower thresholds remain its main errors. These cases motivate explicit start–end interaction and candidate-level boundary refinement.

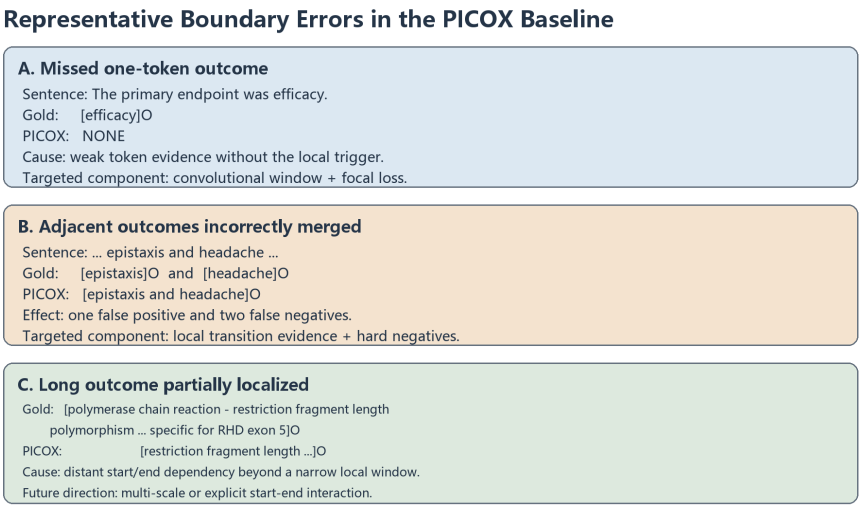


Fig. 4. Representative PICOX failures and their status under the proposed design: (A) a targeted short-span omission, (B) a partially addressed adjacent merge, and (C) an unresolved long-span boundary error.

The Chinese corpus introduces additional limitations. It is substantially smaller than EBM-NLP and is derived from normalized recommendation records, whose recurring templates may make the test distribution more regular than unrestricted guideline prose. The current records contain no explicit control field, so comparison extraction is not evaluated independently. Moreover, structured-field alignment is less costly than annotation from raw documents but may omit PICO expressions not represented by the source schema. Results should therefore be interpreted as evidence of transfer to Chinese recommendation sentences rather than complete Chinese guideline-document extraction.

These observations lead to targeted rather than generic future work. The 16.4% near-boundary errors motivate a lightweight boundary-refinement de-

coder that reranks positions within a two-token neighborhood. The 78.9% short-span share among zero-overlap misses motivates direct enumeration of one- to three-token candidates as a recall backoff, followed by the existing span classifier. For the residual long-span errors, parallel kernels of sizes 3, 5, and 7 or an explicit start–end interaction layer can model wider dependencies. Finally, the 61.2% unseen-string rate motivates terminology-aware domain-adaptive pretraining rather than relying only on memorized span strings. For Chinese guidelines, evaluation will be extended to raw paragraphs with explicit comparison annotations and recommendations from additional diseases and institutions.

5 Conclusion

This paper presents a multi-channel gated feature fusion model for overlapping PICO entity extraction. The model augments PubMedBERT token representations with an explicit local convolutional channel and a global sentence channel. A token-wise softmax gate adaptively combines the three views while preserving the original hidden dimension. Focal loss addresses the imbalance of boundary labels, and validation-based start and end thresholds improve boundary recall. On EBM-NLP, the complete method achieves 58.56% micro F1 and 60.38% macro F1, improving PICOX by 7.69 and 7.61 percentage points and exceeding the Qwen2.5-72B-Instruct three-shot baseline by 4.41 micro-F1 points. On the constructed Chinese clinical guideline corpus, ERNIE-Health+Ours achieves 62.37% micro F1, 6.53 points above ERNIE-Health+PICOX and 3.19 points above the Qwen three-shot baseline. The consistent gains across English RCT abstracts and Chinese guideline recommendations indicate that the model effectively reduces the first-stage missed-boundary problem and is adaptable to different biomedical languages and genres. Guided by the error audit, future work will prioritize two-token boundary refinement for near misses, exhaustive short-candidate backoff for one- to three-token entities, and wider start–end interaction for long spans. The Chinese evaluation will be extended from normalized recommendation sentences to complete guideline paragraphs with explicit comparison annotations.

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References

1. Nye, B., et al.: A corpus with multi-level annotations of patients, interventions and outcomes to support language processing for medical literature. In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics, pp. 197–207 (2018)
2. Lee, J., et al.: BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Bioinformatics **36**(4), 1234–1240 (2020)

3. Beltagy, I., Lo, K., Cohan, A.: SciBERT: a pretrained language model for scientific text. In: Proceedings of EMNLP-IJCNLP, pp. 3615–3620 (2019)
4. Gu, Y., Tinn, R., Cheng, H., et al.: Domain-specific language model pretraining for biomedical natural language processing. ACM Transactions on Computing for Healthcare **3**(1), 1–23 (2021)
5. Kanakarajan, K.R., Kundumani, B., Sankarasubbu, M.: BioELECTRA: pretrained biomedical text encoder using discriminators. In: Proceedings of BioNLP, pp. 143–154 (2021)
6. Lin, T.-Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: Proceedings of the IEEE International Conference on Computer Vision, pp. 2980–2988 (2017)
7. Zhang, N., et al.: ERNIE-Health: A Chinese pre-trained language model for medicine. arXiv preprint arXiv:2110.07244 (2021)
8. Zhang, G., Zhou, Y., Hu, Y., Xu, H., Weng, C., Peng, Y.: A span-based model for extracting overlapping PICO entities from randomized controlled trial publications. Journal of the American Medical Informatics Association **31**(5), 1163–1171 (2024). https://doi.org/10.1093/jamia/ocae065
9. Ghosh, M., Mukherjee, S., Ganguly, A., Basuchowdhuri, P., Naskar, S.K., Ganguly, D.: AlpaPICO: Extraction of PICO frames from clinical trial documents using LLMs. Methods **226**, 78–88 (2024). https://doi.org/10.1016/j.ymeth.2024.04.005
10. Reason, T., Langham, J., Gimblett, A.: Automated mass extraction of over 680,000 PICOs from clinical study abstracts using generative AI: A proof-of-concept study. Pharmaceutical Medicine **38**, 365–372 (2024). https://doi.org/10.1007/s40290-024-00539-6
11. Du, X., Zhao, H., Xing, D., Jia, Y., Zan, H.: MRC-based nested medical NER with co-prediction and adaptive pre-training. In: Proceedings of LREC-COLING, pp. 11669–11679 (2024). https://aclanthology.org/2024.lrec-main.1019/
12. Kim, H., Kim, J.-E., Kim, H.: Exploring nested named entity recognition with large language models: Methods, challenges, and insights. In: Proceedings of EMNLP, pp. 8653–8670 (2024). https://doi.org/10.18653/v1/2024.emnlp-main.492
13. Yun, H.S., Pogrebitskiy, D., Marshall, I.J., Wallace, B.C.: Automatically extracting numerical results from randomized controlled trials with large language models. In: Proceedings of the 9th Machine Learning for Healthcare Conference. PMLR, vol. 252, pp. 1–23 (2024). https://proceedings.mlr.press/v252/yun24a.html
14. Wang, Z., Cao, L., Danek, B., Jin, Q., Lu, Z., Sun, J.: Accelerating clinical evidence synthesis with large language models. npj Digital Medicine **8**, 509 (2025). https://doi.org/10.1038/s41746-025-01840-7
15. Qwen, Yang, A., Yang, B., Zhang, B., Hui, B., et al.: Qwen2.5 technical report. arXiv preprint arXiv:2412.15115 (2025). https://doi.org/10.48550/arXiv.2412.15115
16. Yisha, Z., Zou, P., Li, S., et al.: Assessing data extraction in randomized clinical trials with large language models. BMC Medical Research Methodology **26**, 33 (2026). https://doi.org/10.1186/s12874-025-02729-5